Policy Spillovers to Biodiversity: The Impact of Clean Water Act Investments on Birds

Anthony Ponce

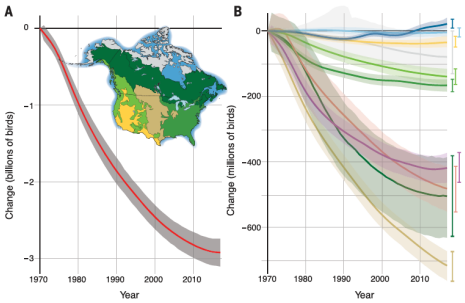
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"Value, in the late Anthropocene, has come almost exclusively to mean utility to human beings. Wild birds, in themselves, will never pull their weight in the human economy. Although some species have been lost forever, their world is still very much alive." - **Jonathan Franzen, National Geographic 2018**

Biodiversity on the Last 50 Years

- **Lost**: about 2.9 billion breeding birds since 1970, 29% of the 1970 total (Rosenberg et al., 2019)
- **Not all**: waterfowl +56% and wetland breeders +13%; waterbirds −22% and shorebirds −37%
- **Functional extinction**: India's vultures gone, human deaths +4.7% (Frank and Sudarshan, 2024)
- **A wintering waterbird**: Chesapeake canvasbacks lost their submerged-plant diet as the beds declined (Perry and Deller, 1996)



From Rosenberg et al. (2019)

Literature on Valuing Biodiversity

- **Recreation demand**: 96 million U.S. birders spent $107.6 billion in 2022 (Carver, 2024)
- **Species carry the value**: one more expected species is worth $3.38 a trip (Kolstoe and Cameron, 2017)
- **Ecosystem services**: waterbirds carry seeds and nutrients between waters (Green and Elmberg, 2014)
- **The gap**: water-policy benefits "abstract from ecological consequences" (Keiser and Shapiro, 2019b)

▸ Services by Species



"By year's end, they had logged 38,757 miles" on "a $16,000 budget" and "identified 579 bird species" (Woodbury, 2025)

Policy Setting of the CWA

- **Title II**: 75% of a treatment plant's cost, nearly $52 billion in grants, 1973 to 1990 (Ramseur, 2023)
- **By rule**: projects ranked on pollution severity and people served, never on wildlife (40 CFR 35.915)
- **Timing**: 40% of major municipal dischargers met the 1977 deadline (U.S. General Accounting Office, 1980)
- **What varies**: when the first plant came online, and its share of the river

▸ Treatment Technologies



Cuyahoga River fire, 1969

Research Question

What are the impacts of federal investments in public infrastructure on ecological changes in biodiversity[1]?

1. **Extensive:** How does grant-funded sanitation plant construction, its commissioning[2], under the CWA's Title II program affect bird biodiversity?

2. **Intensive:** Are changes in biodiversity downstream of a treatment plant linked to that plant's grant-funded contributions of effluent treated flow-share improvements?

[1] biodiversity: richness (number of different species), abundance (number of a given species)
[2] commissioning: a plant's earliest completion of a funded Step 3, 4, 7, or 9 construction grant

Preview of Results

Commissioning: 5% more water-linked species per survey (about 1 at the average survey, 1 every 2 at the median) and, with the largest flocks dropped, 14% more water-linked birds (830 at the average survey, 85 at the median)

Against no commissioning, 1967 to 2024: 3% more water-linked species (870 species entries on surveys' lists a year) and, with the largest flocks dropped, 9% more water-linked birds (0.7 million birds counted a year)

The dose: circles with a higher dose at commissioning gained no more than circles with a lower one, but each added plant upstream brings about 1% more water-linked species per survey

Pathway: dissolved oxygen rose and phosphorus fell after commissioning, but the data do not identify how much of the birds' gain operates through water quality

Contributions To Literature

Ecosystem-service valuation starts from a measured ecological outcome and a stated link to the people who benefit (Daily et al., 2009; Boyd and Banzhaf, 2007; Olander et al., 2018)

Related literature: air pollution rules (Liang et al., 2020), shale wells and wind turbines (Katovich, 2024), forest infrastructure (Madhok, 2025)

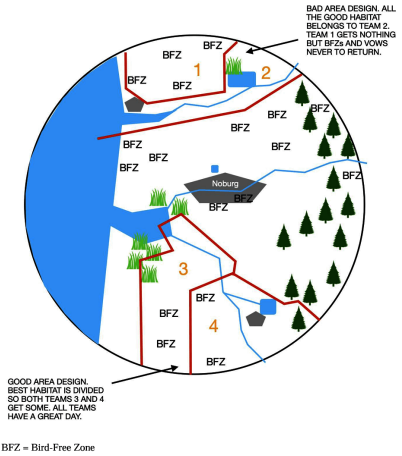
This paper: fills gaps in ecology and policy valuation by quantifying impacts of spillovers to biodiversity through investments in water quality technologies.

How: Causal models pairing variation in CWA grant allotment & exogeneity of hydrogeography with novel, historical biodiversity data

Audubon Society's Christmas Bird Count (CBC) Raw Data

As of the 125th CBC (2024-2025), over 100k reports archiving sightings for over 1k different species of birds

- Counts maintained to circles of 24-kilometer diameters ▸ Counts per Survey
- Teams of birders go out to count as many birds as possible led by 1-2 organizers ▸ Survey Effort
- Databases and papers for guild and conservation assignments (De Graaf et al., 1985; Wilman et al., 2014; Tobias et al., 2022)
- **Water-linked species**: species that forage on or in the water, live in wetlands, or eat a food whose supply shifts with water quality ▸ Water-Linked Species



Title II Grants, Plant Characteristics, & Hydrology

Treatment: EPA **Grants Information and Control System** grants (Keiser and Shapiro, 2019a)

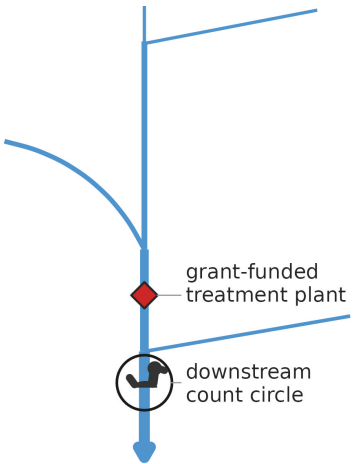
▸ Grants over Time

CWNS plant characteristics: (commissioning date, design flow, treatment level, more) (Jerch, 2026)

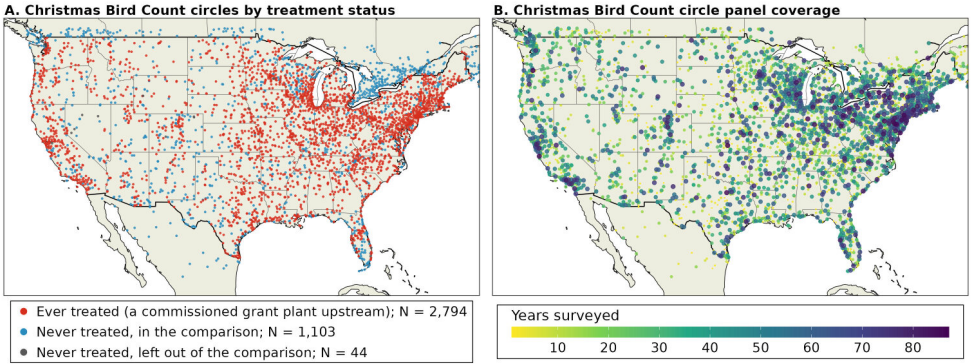
▸ Design Flows

Network: NHDPlus Version 2.1 (McKay et al., 2012) - 25-mile upstream-downstream networks map plants to circles

Controls:
water quality (Water Quality Portal, 2024) winter climate from ERA5 (single levels, monthly)
housing-unit density (Markley et al., 2022)
land in farms and harvested cropland (Haines et al., 2014)

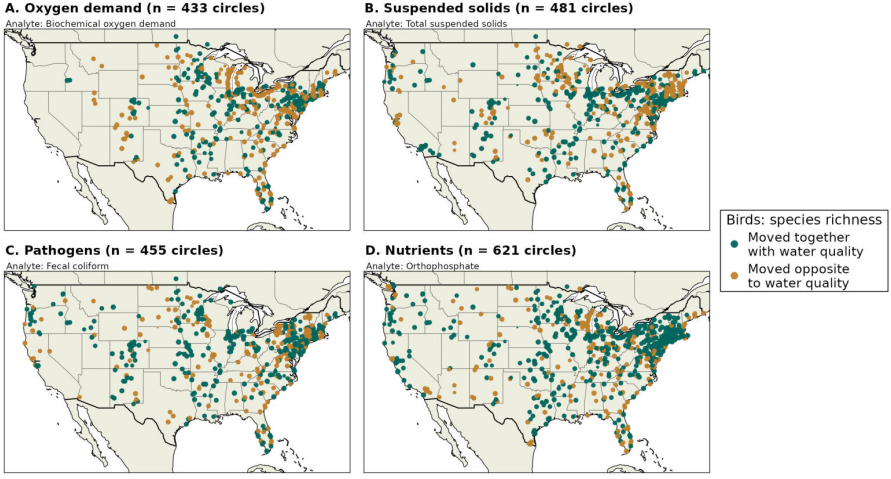


Constructed Panel



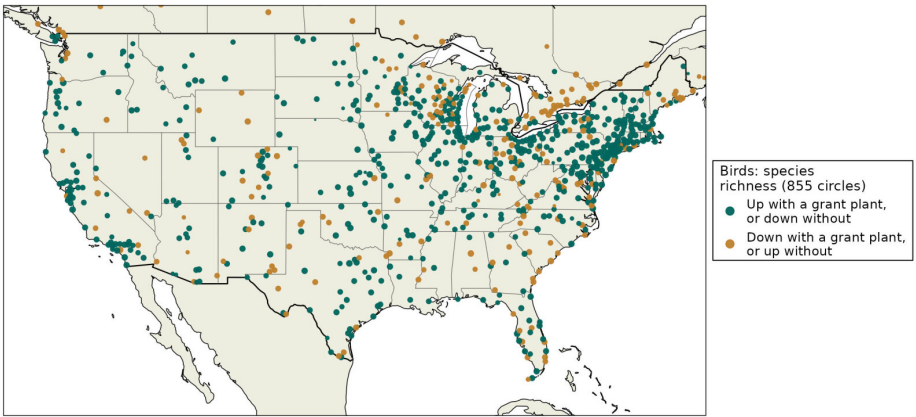
▸ How the Panel Is Built

Motivating Descriptives - Water Quality



Long Differences (Hsiang, 2016) in Birds Overlayed with Long Differences in Water, Species Richness

Motivating Descriptives - Grants



Long Differences (Hsiang, 2016) in Birds against Grant-Plant Commissioning, Species Richness
Windows 1951-1972 and 2000-2024. A grant plant is within 25 river miles upstream and first commissioned between them.

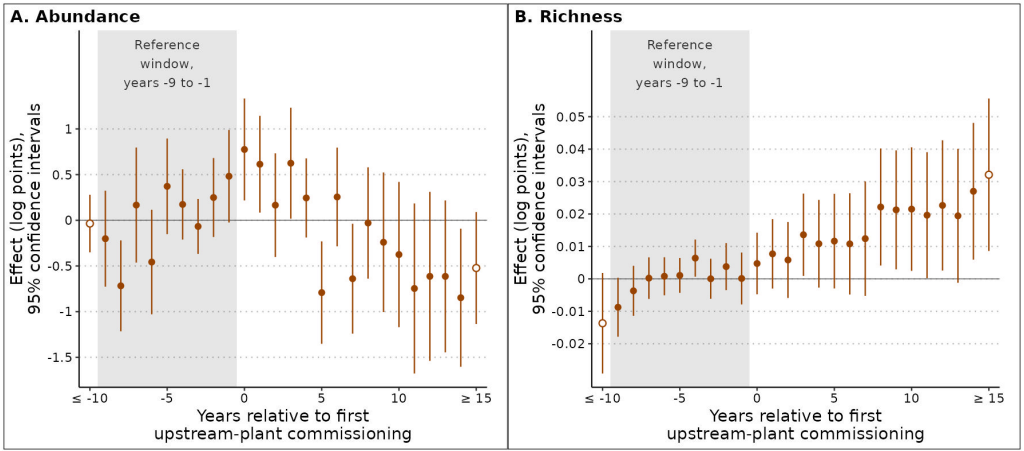
How does grant-funded construction affect biodiversity?

$$\text{Poisson:}\quad \underbrace{\mathbb{E}[\,N_{ct} \mid \cdot\,]}_{\substack{\text{expected count,}\\ \text{given the circle}}} = \underbrace{\phi_c}_{\substack{\text{circle effect,}\\ \text{conditioned out}}} \exp\Bigg(\underbrace{\lambda_t}_{\text{year FE}} + \sum_{k \neq -1} \underbrace{\beta_k}_{\substack{\text{effect } k \text{ years}\\ \text{after commisioning}}} \underbrace{\mathbf{1}[\,t - G_c = k\,]}_{\substack{\text{event-time}\\ \text{dummy}}} + \underbrace{\mathcal{C}'_{ct}\,\delta + \mathcal{T}_{ct}}_{\substack{\text{controls,}\\ \text{baseline trends}}} \Bigg) \tag{1}$$

where,

- N_{ct}: abundance or richness on circle c's survey in year t, with c the **CBC circle** and t the **count year**
- $\phi_c = \exp(\alpha_c)$: the circle fixed effect as a *multiplier*
- G_c: the year the circle's first upstream plant *within 25 river-miles* is commissioned
- β_k: the effect k CBC years from that grant, *against the the average pre-treatment years*.
- $\mathcal{C}_{ct}$: survey effort (party-hours, counters, miles) and winter temperature and precipitation
- $\mathcal{T}_{ct}$: the circle's baseline housing density and cropland share, each interacted with year

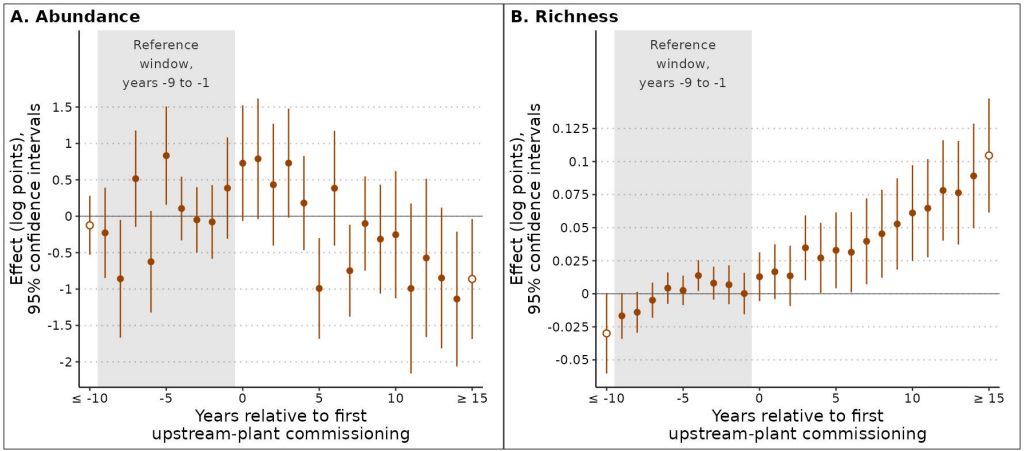
Extensive Margin - Across All Birds



Standard errors clustered at the circle level. Reference period: de-meaned trend in pre-treatment coefficients (Miller, 2023).

▸ Specification Chart

Extensive Margin - Water-linked Birds



Standard errors clustered at the circle level. Reference period: de-meaned trend in pre-treatment coefficients (Miller, 2023).

▸ Specification Chart

Key Takeaways - Effects of Grant-Funded Commissioning

Richness: 5% (about 1 at the average survey/ 1 every 2 at the median survey) more water-linked species per survey, and 5 to 8% across the water-linked groups with statistically significant gains

Abundance: 14% (830 at the average survey/ 85 at the median survey) more water-linked birds per survey with the largest flocks dropped, and up to 42% for submerged-plant grazers on the full counts

Checks for Causality on Grant Commissioning

Concerns for endogeneity:

1. **Targeting**: grants went to the worst waters and largest towns, when they did
2. **Interference (SUTVA)**: a circle's birds may answer to plants that are not its own, through the water or by moving
3. **Measurement**: people make the counts, and their effort and coverage are not fixed

Checks:

1. **Targeting**: pre-program birds do not predict when a plant came (p-values 0.23 to 0.83). Circles with fewer species were likelier to receive treatment. ▸ The Test
2. **Interference**: untreated circles near treated ones match those beyond 200 km (p-values 0.24 to 0.68). No sign of birds drawn from neighbors ▸ The Test
3. **Measurement**: no jump in effort at commissioning (p-value 0.51), and circles do not quit after poor years. Effort rises after rich counts ▸ The Test

SUTVA: the stable unit treatment value assumption, that one circle's treatment does not move another circle's birds (Butts, 2026).

Simulation

$$\underbrace{\text{Gain}}_{\text{a year}} = \underbrace{\frac{1}{T}\sum_{t=1967}^{2024}}_{\text{the average year, first commissioning to 2024}} \underbrace{\sum_{c:\ t \geq G_c}}_{\text{treated surveys at or after commissioning}} \underbrace{N_{ct}}_{\text{species or birds recorded}} \underbrace{\left(1 - e^{-\hat{b}}\right)}_{\text{the share due to commissioning, } \hat{b} \text{ the post-period mean (0.045 for water-linked species)}}$$

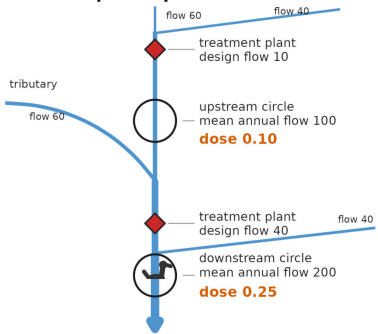
$$\underbrace{\text{Continent-wide gain}}_{\text{an upper bound}} = \underbrace{P_{2017}}_{\text{population (Rosenberg et al., 2019)}} \times \underbrace{s_{2008\text{-}2017}}_{\text{share of the group's birds counted in treated circles, 2008 to 2017}} \times \underbrace{\left(1 - e^{-\hat{b}}\right)}_{\text{the share due to commissioning}}$$

Against no commissioning, 1967 to 2024 (Liang et al., 2020):

- 3% (870 species entries on surveys' lists a year) more water-linked species, and 3 to 5% across those groups
- 9% (0.7 million birds counted a year) more water-linked birds with the largest flocks dropped, and up to 25% for submerged-plant grazers on the full counts
- Continent-wide upper bounds: 6 to 65 million birds (Rosenberg et al., 2019)

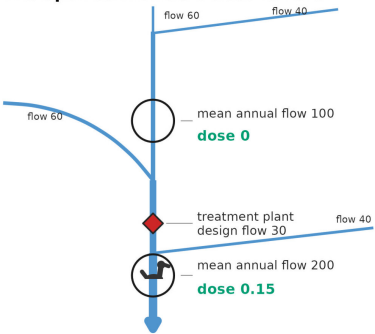
Visualization

A. A second plant upstream doses both circles



dose difference +0.15

B. The upstream circle's dose is zero



dose difference +0.15

Policy Mechanisms

Dosed Treatment = staggered commissioning × dilution of the **effluent treated flow share** FS from plant p to circle c:

$$\text{FS}_{pc} = \frac{\text{plant design flow}}{\text{mean annual flow at the outlet}} \qquad D_{ct} = \sum_{p} \text{FS}_{pc} \cdot \mathbf{1}[\text{commissioned}_p \leq t]$$

- FS: *the fraction of the circle's water that passed through the plant.* A grant event becomes a measured, **continuous** dose (Callaway et al., 2025), comparable across networks of tributaries (Fan, 2025).
- **Identification**:
  - **PT** ⇒ circles that ended at different doses on a common trajectory beforehand
  - ***Strong* PT** ⇒ each circle would have followed that same trajectory *at any other dose* (Callaway et al., 2025)

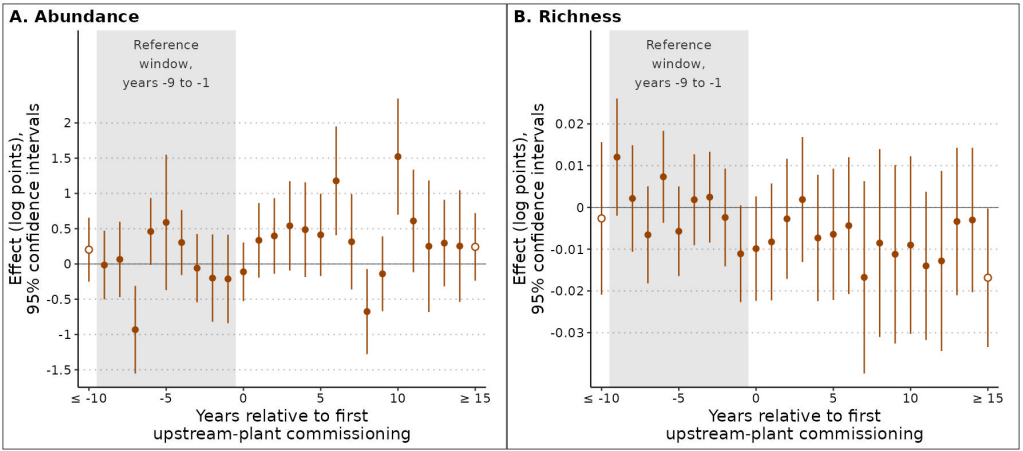
Are changes in biodiversity linked to effluent treated flow-share?

Median Split: $$\mathbb{E}[N_{ct} \mid \cdot] = \phi_c \exp\Bigg(\underbrace{\lambda_{G_c t}}_{\substack{\text{cohort-by-}\\ \text{year FE}}} + \sum_{k \neq -1} \underbrace{\psi_k}_{\substack{\text{high v. low}\\ \text{at event}\\ \text{time } k}} \underbrace{\text{High}_c}_{\substack{\text{above}\\ \text{cohort's}\\ \text{median dose}}} \mathbf{1}[t - G_c = k] + \mathcal{C}'_{ct}\delta + \mathcal{T}_{ct} \Bigg) \quad (2)$$

Dynamics of dose change: $$\mathbb{E}[\, N_{ct} \mid \cdot \,] = \phi_c \exp\Bigg(\lambda_t + \sum_{\tau \neq -1} \underbrace{\varphi_\tau}_{\substack{\text{level response/dose}\\ \tau \text{ later}}} \Delta D_{c,t-\tau} + \mathcal{C}'_{ct}\,\delta + \mathcal{T}_{ct} \Bigg) \quad (3)$$

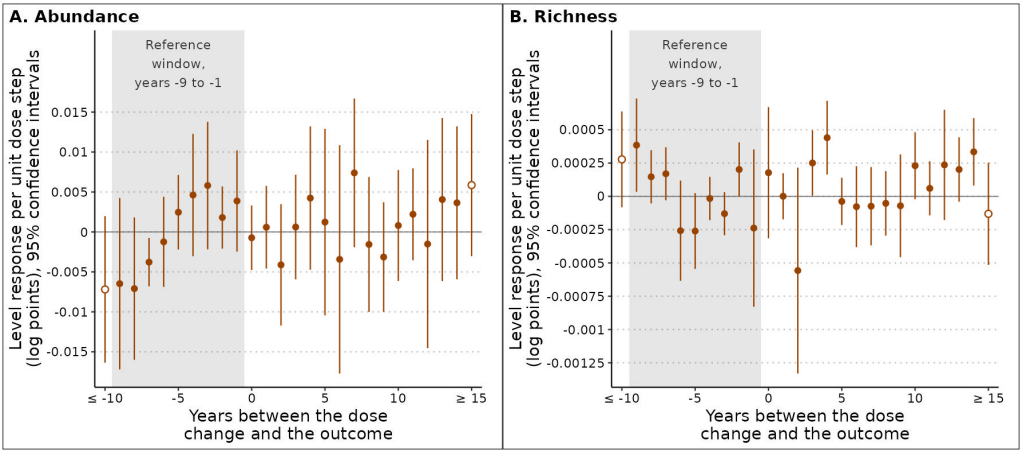
- $\text{High}_c = 1$ if the circle's dose at commissioning is above the *median among circles commissioning in the same year*
- $\lambda_{G_c t}$ controls comparisons of high against low *within* that cohort
- $\Delta D_{c,t-\tau}$ = the change in the circle's dose τ years earlier
- φ_τ = the proportional change in the circle's birds τ years after its dose rises

Results - Relative to Median



Outcome: All birds. Standard errors clustered at the circle level. Reference period: de-meaned trend in pre-treatment coefficients (Miller, 2023)

Results - Size of Dose Change



Outcome: All birds. Standard errors clustered at the circle level. Reference period: de-meaned trend in pre-treatment coefficients (Miller, 2023)

Key Takeaways - Effects of Dose

Null dose: circles with a higher dose at commissioning gained no more species or diversity than circles with a lower one

No dose-response slope: responses to dose steps fail their pre-trend tests in 76 of 84 fits, and a circle's birds before the program predict the dose it later received

Number of plants: each added plant upstream brings about 1% more water-linked species per survey fourteen years on

▸ The Regression

Checks for Causality on Dose

Concerns for endogeneity:

1. **Selection on gains**: design flow is chosen for a town's growth, where birds' trends may differ
2. **Technology**: ponds and lagoons sit on cheap, flat land beside wetlands, where birds are many
3. **The plant as a place**: ponds and ice-free water at a plant can draw birds and observers

Checks:

1. **Selection on gains**: circles taking a larger share of the same plant's flow gain no more birds (p-value 0.64). Circles with more birds before the program received larger doses, 72% more per standard deviation (joint p-value below 0.001) ▸ The Test
2. **Technology**: pre-program birds do not predict a pond plant upstream (p-values 0.09 or above). Commissionings that only added capacity still brought 97% more birds per survey ▸ The Test
3. **The plant as a place**: each plant whose effluent misses the circle's outlet brings 36% more birds per survey ▸ The Test

Baseline Improvements

Pathway: dissolved oxygen rose and phosphorus fell after commissioning

▸ Water Quality

Suspended solids and fecal coliform respond to the dose differently below dirtier and cleaner plants. Different effects on birds (Y_{ct})?

$$Y_{ct} = \alpha_c + \lambda_t + \underbrace{\beta^D D_{ct}}_{} + \underbrace{\beta^P \hat{P}_{ct}}_{\text{the effect through predicted cleaning}} + \mathcal{C}'_{ct}\delta + \mathcal{T}_{ct} + \varepsilon_{ct}, \qquad \hat{P}_{ct} = \underbrace{\hat{\beta}^M_{h(c)}}_{\text{analyte's response to the dose}} D_{ct}$$

$\hat{\beta}^M_{h(c)}$: circle + year effects, split at the median within each commissioning cohort (as in Equation 3)

- Suspended solids: dose's coefficient is unchanged (0.0063 against 0.0056)
- Fecal coliform: predicted cleaning carries effect (0.61 SD per SD, SE 0.24) - the dose's own coefficient falls to zero
- Both fail their pre-trend tests in event time - impact on birds through the baseline water improved is not identified

▸ Evidence

Recap of Results

Commissioning: 5% more water-linked species per survey (about 1 at the average survey, 1 every 2 at the median) and, with the largest flocks dropped, 14% more water-linked birds (830 at the average survey, 85 at the median)

Against no commissioning, 1967 to 2024: 3% more water-linked species (870 species entries on surveys' lists a year) and, with the largest flocks dropped, 9% more water-linked birds (0.7 million birds counted a year)

The dose: circles with a higher dose at commissioning gained no more than circles with a lower one, but each added plant upstream brings about 1% more water-linked species per survey

Pathway: dissolved oxygen rose and phosphorus fell after commissioning, but the data do not identify how much of the birds' gain operates through water quality

Policy Application

Effects of grants: more species counted below the plants, mostly water-linked

Extensive v Intensive: effects from plants, not their share of contributions to cleaning - siting matter at least as much as dilution

CWSRF: the dose cannot yet rank sites, so which plants to fund, and where, is the targeting question the research agenda takes up next

Benefits and costs: Office of Management and Budget's November 2023 Circulars A-4 and A-94

Non-monetized Benefits

	Monetized, billions of 2014 dollars		**Non-monetized**, water-linked species		
	Housing values	Infant health	Birds per survey	Species per survey	Entries a year
Benefits average, median	89	in the ratio	+14% 830 and 85	+5% 1 and 1 every 2	+3% 870 against no grants
Costs	341 present value, 650 to 870 over the projects' lives				
Benefits over costs	0.24 to 0.26	up to 0.45, with housing			
Source	Keiser and Shapiro (2019a), Keiser et al. (2019)	Flynn and Smith (2022)			
	Not quantified				
	Ecosystems beyond the birds	Nonuse values	Recreation beyond 25 miles	Fisheries and water supplies	

Dollars: present values at 7.85% over 30 years (0.26), annual flows over 1962 to 2001 (0.24), and lifetime totals in 2014 and 2017 dollars. The health ratio adds infant health (Flynn and Marcus, 2025) to housing values. Gains apply each percent to treated circles' surveys in the nine years before commissioning (average 18 species and 6,021 birds, median 13 and 620, extreme counts dropped). Entries: one species on one survey's list.

Pipeline For Research Agenda

- **Targeting of grants** - generalized random forests (Athey et al., 2019): what matters most for investments and where do they matter most for biodiversity?
- **Trophic gradients** - specialist to generalist compositions: at the individual-bird level, why do some birds benefit more than others? (Root, 1967; Julliard et al., 2006; Karr, 1981; Şekercioğlu, 2006)
- **From measurement to monetization** - using existing tools for recreation demand (Kolstoe and Cameron, 2017; Melstrom and Reeling, 2026; Wardle, 2025)

Thank You



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Appendix - Ecosystem Services by Species

Species	Feeding guild	Ecosystem services of its group	Sources
Mallard	Submerged-plant grazer	Hunting, seed dispersal to new and restored wetlands, early detection of avian influenza	Green and Elmberg (2014), U.S. Fish and Wildlife Service (2023)
Canvasback	Submerged-plant grazer	Hunting, grazing that can raise submerged-plant diversity, a diet that tracked the loss of submerged plants	Green and Elmberg (2014), Perry and Deller (1996)
Lesser Scaup	Bottom-sediment feeder	Hunting, predation that reduces invasive zebra mussels, whose U.S. damages and control cost about $1 billion a year	Green and Elmberg (2014), U.S. Fish and Wildlife Service (2023)
Ring-billed Gull	Scavenging gull	Disposal of waste at dumps, removal of nutrients from dumps and treatment plants, control of farm pests	Green and Elmberg (2014)
Great Blue Heron	Visual fish-eater	Removal of fish before die-offs as waters fall, predation on crayfish pests, a receptor species for risk assessments of contaminants	Green and Elmberg (2014), Seston et al. (2009)

The paper classifies each water-linked species into one feeding guild. The review reports services by group (dabbling and diving ducks, gulls, herons), several as likely and mostly from studies outside North America. None is measured for these species here.

Back

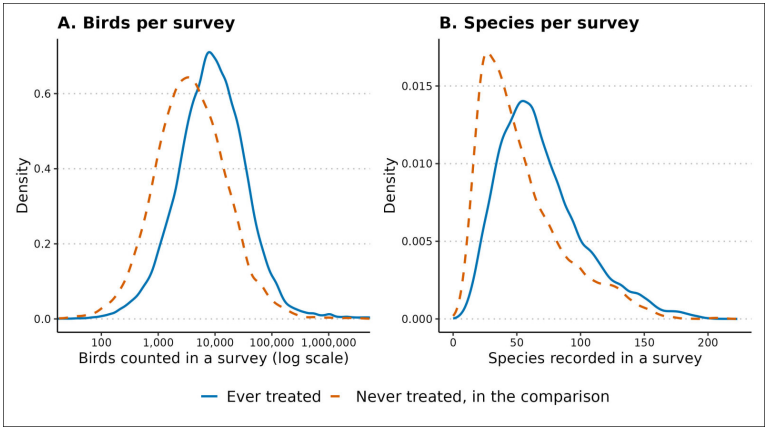
Appendix - Treatment Technologies, Costs, and Birds

Technology	Category	How it works	Lifetime	Lifetime cost	Benefits for birds
Facultative lagoon	I, secondary	Basins 8 to 16 feet deep in which aeration aids digestion	20 years	$0.42, Aroostook-Prestile, ME	Oxygen rises, plants and invertebrates recolonize, and birds may benefit (Pringle and Burton, 2017)
Biotower	I, secondary	Bacteria grown on a plastic medium degrade the organic matter	20 years	$0.39, Tolleson, AZ	
Activated sludge with a pressure filter	I, secondary	Bacteria degrade the organic matter. The filter, added in case limits tightened, proved unnecessary	20 years	$1.01, Valley Forge, PA, grant-funded	
Nitrification and phosphorus removal	IIB, advanced	Bacteria convert ammonia to nitrate, and filters remove phosphorus	–	–	Submerged plants, food for grazing waterbirds, returned to the upper Potomac in 1983, with better treatment and unusual weather (Dennison et al., 1993)
Constructed wetland	Innovative or alternative	Effluent flows through a planted wetland	–	$166,000 to build, against $500,000 to upgrade conventionally, Monterey, VA	Habitat, with 412 bird species recorded at 28 treatment wetlands (Knight et al., 2001)

Lifetime cost: capital plus operating costs over the assumed 20-year life, discounted at 10%, in 1983 dollars per 1,000 gallons (Congressional Budget Office, 1985). Each category is a class of need in the 1982 Needs Survey (U.S. Environmental Protection Agency, 1982). Advanced treatment: Blue Plains, Washington, D.C. (Carter and Rybicki, 1986; Ruhl and Rybicki, 2010). Wetland costs: U.S. General Accounting Office (1994). – not reported.

Back

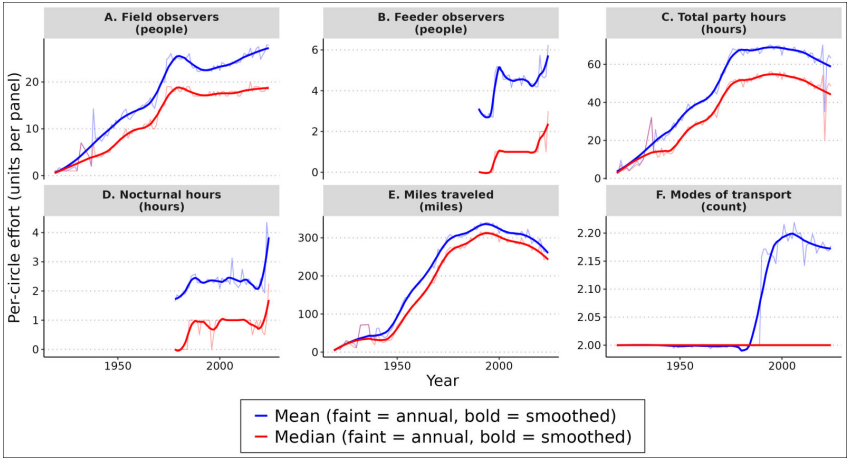
Appendix - Birds and Species per Survey



All species, every survey from 1919 to 2024: 75,485 surveys at 2,794 ever-treated circles and 28,215 at 1,386 never-treated circles in the comparison.

Back

Appendix - Survey Effort over Time



Each year's mean and median of six effort measures over every circle's survey. Effort enters every specification as a control.

Back

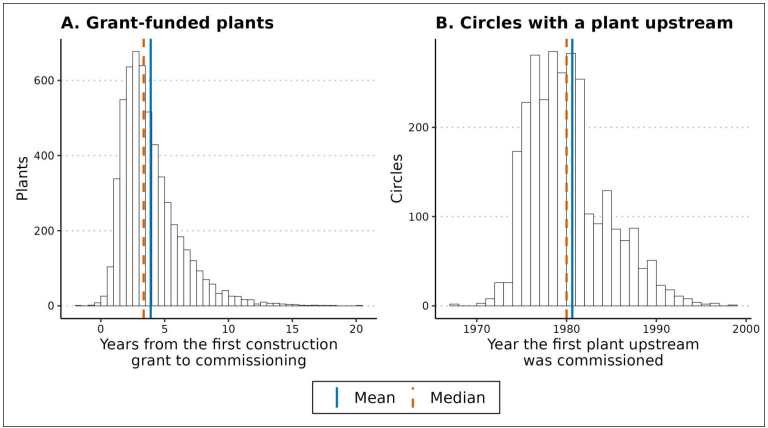
Appendix - Water-Linked Species by Guild and General Group

Feeding guild	Waterbirds	Waterfowl	Shorebirds	Landbirds	Introduced	No group	All
Visual fish-eater	42	2	0	2	0	22	68
Crustacean-eater	6	0	1	0	0	3	10
Aerial insect-eater	0	0	0	6	0	0	6
Marsh invertebrate-eater	13	2	32	10	0	12	69
Submerged-plant grazer	3	15	0	0	1	2	21
Bottom-sediment feeder	1	14	0	1	0	1	17
Scavenging gull	9	0	0	0	0	3	12
Scavenging crow	0	0	0	1	0	0	1
Water-linked	74	33	33	20	1	43	204
Not water-linked	3	6	10	330	7	189	545
All species	77	39	43	350	8	232	749

Species recorded on the count, by feeding guild and by the general groups of Rosenberg et al. (2019). A species is water-linked if it has a feeding guild. No group: species the general groups do not classify.

Back

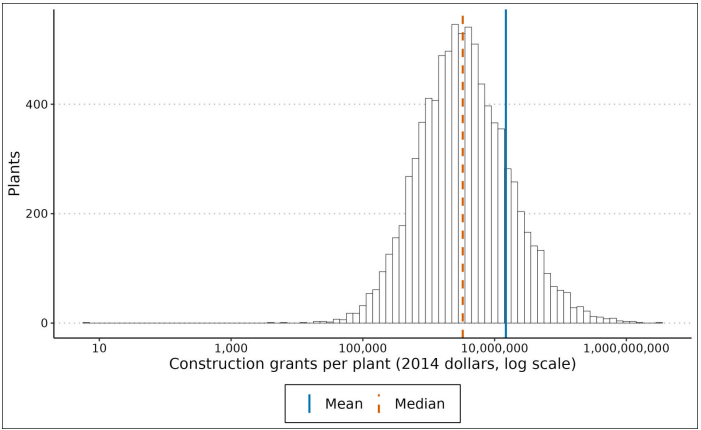
Appendix - Grants over Time



Panel A: the 5,641 grant-funded plants within 25 river miles upstream of a surveyed circle, by the years from their first construction grant to commissioning (median 3.4). Panel B: the 2,794 ever-treated circles, by the year their first plant upstream was commissioned (median 1980).

Grants per Plant Back

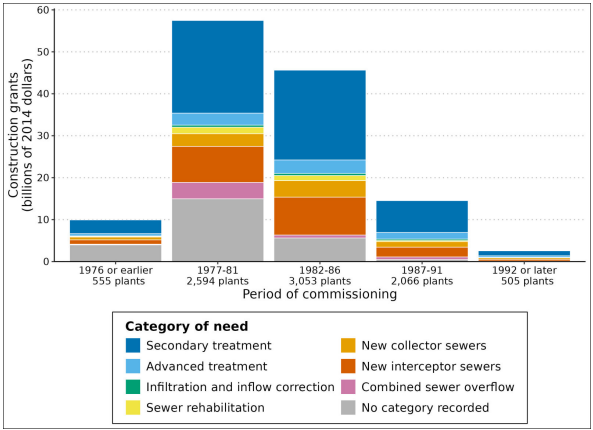
Appendix - Grants per Plant



The construction grants of each of the 8,773 grant-funded plants, in 2014 dollars, each grant deflated by the consumer price index of its award year. The median plant received $3.3 million and the mean $14.8 million, since a few large plants received over $1 billion each.

By Category of Need | By Level of Treatment | By Technology | Back

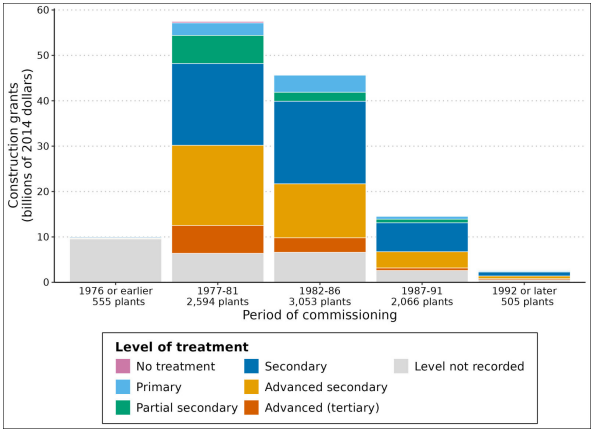
Appendix - Grants by Category of Need



Construction grants by period of commissioning. Each grant's dollars are split across the Needs Survey's categories of need in proportion to its eligible costs. Of the dollars with a category recorded, about three in five paid for treatment and two in five for sewers.

Grants per Plant Back

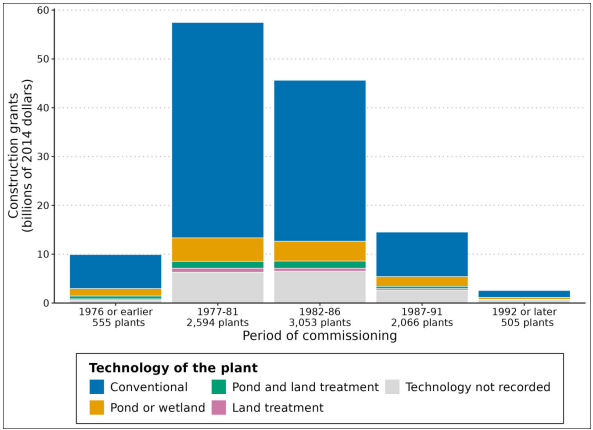
Appendix - Grants by Level of Treatment



Construction grants by period of commissioning and the plant's level of treatment in the Needs Survey wave the design flow is taken from. Most dollars went to plants at secondary treatment or beyond. The 1973 and 1976 waves record no level.

Grants per Plant Back

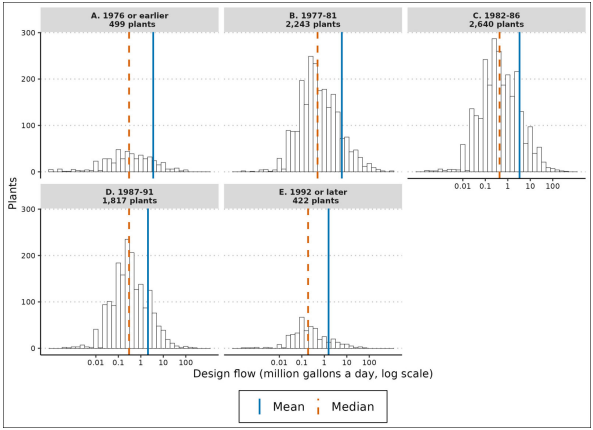
Appendix - Grants by Technology



Construction grants by period of commissioning and the plant's technology, recorded for 7,046 of the 8,773 plants. Conventional plants received about $95 billion of the $130 billion. Ponds, wetlands, and land treatment received about a seventh.

▸ Grants per Plant ◂ Back

Appendix - Design Flows by Period



Design flows of the 7,621 grant-funded plants with one recorded, by period of commissioning, in the Needs Survey wave the dose uses. Over all periods the median plant was designed for 0.40 million gallons a day and the mean for 3.74.

▸ Plants by Period ◂ Back

Appendix - Grant-Funded Plants by Period

	1976 or earlier	1977-81	1982-86	1987-91	1992 or later
Plants	556	2,594	3,053	2,066	505
Years from the first construction grant to commissioning					
Median	2.1	3.1	3.5	3.0	3.9
25th to 75th percentile	1.4 to 3.1	2.1 to 4.3	2.4 to 5.0	2.0 to 4.8	2.6 to 6.0
Share taking more than 5 years (percent)	6.3	16	26	23	35
Share first awarded before October 1984 (percent)	100	100	99	42	13
Median design flow (million gallons a day)	0.30	0.50	0.43	0.30	0.19
Median construction grants (millions of 2014 dollars)	2.1	4.3	3.8	2.4	2.1
Share holding a Step 4, 7, or 9 grant (percent)	15	10	33	47	62
Circles first treated in the period	466	1,341	664	276	47

All 8,774 grant-funded plants, by period of commissioning. Design flow is taken from the same Needs Survey wave as the dose. Grants are each plant's funded construction grants, deflated by the consumer price index of the award year.

Back

Appendix - How the Panel Is Built

	Step	Rule	What it leaves
1	Species	Each CBC name matched to the 2023 eBird taxonomy. A subspecies counts toward its species, a hybrid or a pair toward none	749 species
2	Zero-filling	A species missing from a survey enters as zero when its BirdLife range lies within 100 km of the circle and it was reported in the circle's Bird Conservation Region	5,135 circles in 30 regions
3	Effort	A circle stays if hours were recorded for the observers present in at least one year	105,057 surveys at 4,224 circles, from 1919
4	Commissioning	A plant's earliest completion of a funded Step 3, 4, 7, or 9 construction grant, a median of three years after its earliest award	8,769 plants with a completion date
5	Plants to circles	The river is followed 25 miles downstream of each plant on NHDPlus Version 2.1. A circle is linked to the plant when that stretch of river crosses it	2,794 ever treated, 1,430 never treated
6	Final sample	All three effort measures, winter climate from 1940, baseline land use on half the circle, and more than one year	85,934 surveys at 2,975 circles, 1940 to 2024

Counts at 25 river miles. The 44 never-treated circles that hold a grant plant on one of their own reaches leave the comparison.

Back

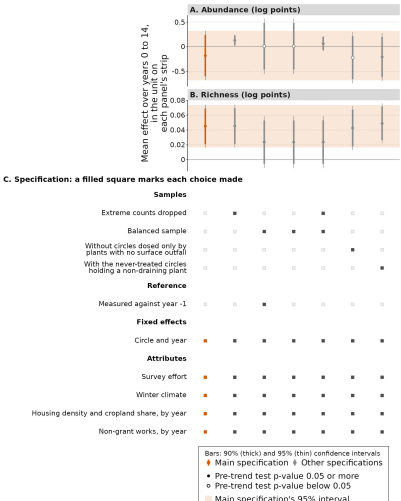
Appendix - Specification Chart, All Birds



Each column is one specification of the event study: its mean effect over years 0 to 14, against the mean of years -9 to -1, after Ortiz-Bobea et al. (2021). The squares beneath the estimates mark each specification's controls and sample. Standard errors clustered at the circle level.

Back

Appendix - Specification Chart, Water-linked Birds



Each column is one specification of the event study: its mean effect over years 0 to 14, against the mean of years -9 to -1, after Ortiz-Bobea et al. (2021). The squares beneath the estimates mark each specification's controls and sample. Standard errors clustered at the circle level.

Back

Appendix - Targeting: When the Plants Came

Question: do a circle's birds before the program predict when, or whether, a grant plant came upstream?

$$\underbrace{Y_c}_{\substack{\text{first commissioning}\\ \text{year, or ever treated}}} = \alpha + \underbrace{\omega_L\, z_c^L}_{\substack{\text{birds' 1951-1972}\\ \text{level, in SD}}} + \underbrace{\omega_T\, z_c^T}_{\substack{\text{their trend, net of}\\ \text{effort, in SD}}} + \underbrace{\delta' \ln E_c^{\text{pre}}}_{\substack{\text{pre-program}\\ \text{effort}}} + \underbrace{\gamma' H_c}_{\substack{\text{largest river's flow}\\ \text{and drainage area}}} + \varepsilon_c \qquad (4)$$

where,

- c: one circle, its surveys over 1951 to 1972 averaged
- The test: $\omega_L = \omega_T = 0$, jointly

Evidence Back

Appendix - Targeting: Evidence

	(1) Ever treated	(2) First commissioning year	(3) Top priority-list position
Panel A. The circle's pre-program abundance			
Level, per standard deviation	−0.018	0.031	−0.025
	(0.018)	(0.27)	(0.034)
Trend, per standard deviation	−0.012	0.23	−0.0027
	(0.013)	(0.17)	(0.024)
Joint p-value, level and trend	0.33	0.42	0.73
Panel B. The circle's pre-program richness			
Level, per standard deviation	−0.062***	−0.060	−0.078**
	(0.024)	(0.32)	(0.038)
Trend, per standard deviation	−0.015	0.23	−0.037
	(0.014)	(0.20)	(0.028)
Joint p-value, level and trend	0.031	0.51	0.068
Circles	536	474	406
Pre-program effort, largest river's flow and drainage area	Yes	Yes	Yes

Least squares, one row per circle, (1) and (3) as linear probabilities. Robust standard errors in parentheses. ***, **: p-values below 0.01, 0.05. Birds do not predict when a plant came. Circles with fewer species were likelier to be treated, a level the circle effects absorb.

Back

Appendix - Interference: Birds Drawn From Neighbors

Question: do surveys at circles with no plant upstream differ by the circle's distance to the nearest treated circle?

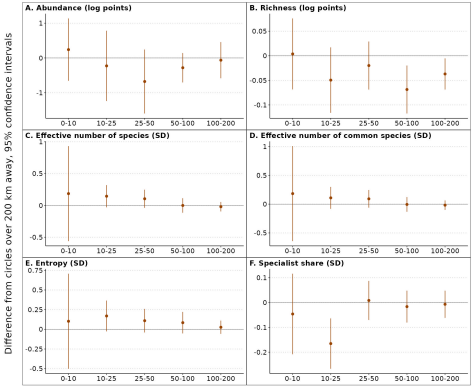
$$\mathbb{E}[\, N_{ct} \mid \cdot \,] \;=\; \phi_c \exp\Bigg(\lambda_t \;+\; \sum_b \underbrace{\mu_b}_{\substack{\text{difference from circles} \\ \text{over 200 km away}}} \underbrace{\mathbf{1}[\, d_{ct} \in b \,]}_{\substack{\text{distance bin to the} \\ \text{nearest treated circle}}} \Bigg) \tag{5}$$

where,

- Sample: surveys at circles with no plant upstream
- Bins: 0-10, 10-25, 25-50, 50-100, and 100-200 km
- The test: $\mu_{\text{0-10}} = \mu_{\text{10-25}} = 0$ (Butts, 2026)

Evidence Back

Appendix - Interference: Evidence



Circle and year effects, standard errors clustered by circle. The bins within 25 km are jointly zero for abundance, richness, and the diversity statistics (p-values 0.24 to 0.68). The specialist share differs at 10 to 25 km.

Back

Appendix - Measurement: Effort at Commissioning

Question: does survey effort jump when the first plant upstream is commissioned?

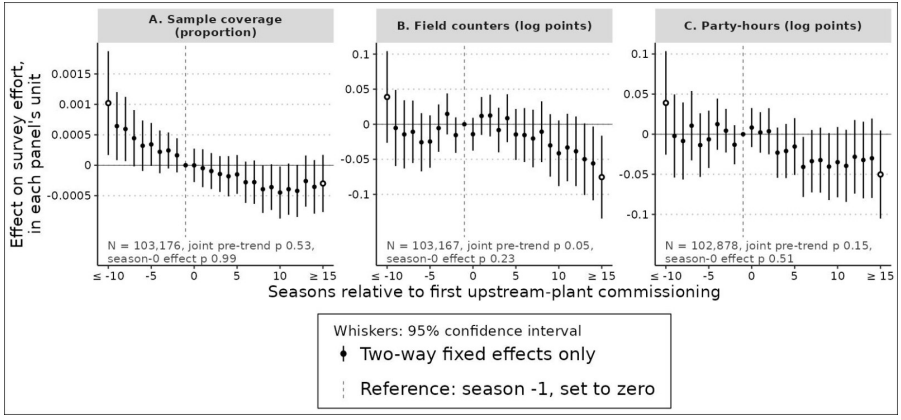
$$\underbrace{E_{ct}}_{\substack{\text{log party-hours, log field} \\ \text{observers, or sample coverage}}} = \alpha_c + \lambda_t + \sum_{k \neq -1} \underbrace{\beta_k}_{\substack{\text{change in effort} \\ k \text{ years after}}} \mathbf{1}[\, t - G_c = k \,] + u_{ct} \tag{6}$$

where,

- Circle and year effects only, the year before commissioning set to zero
- The test: $\beta_0 = 0$, no jump in the year of commissioning

▸ Evidence ◂ Back

Appendix - Measurement: Evidence



Effort does not jump in the year of commissioning (p-values 0.23 to 0.99). Party-hours and field observers decline over the later years, and effort rises after a rich count, which weakens the richness pre-trend test.

Back

Appendix - Selection on Gains: The Dose

Question: do a circle's birds before the program predict the dose it later received?

$$\underbrace{\ln D_c^{\mathrm{T}}}_{\text{log of the dose the circle ends with}} = \alpha + \underbrace{\omega_L\, z_c^L}_{\text{birds' 1951-1972 level, in SD}} + \underbrace{\omega_T\, z_c^T}_{\text{their trend, net of effort, in SD}} + \underbrace{\delta' \ln E_c^{\mathrm{pre}}}_{\text{pre-program effort}} + \underbrace{\gamma' H_c}_{\text{largest river's flow and drainage area}} + \varepsilon_c \qquad (7)$$

where,

- c: one treated circle, its surveys over 1951 to 1972 averaged
- The test: $\omega_L = \omega_T = 0$, jointly. The same right-hand side predicts the grants upstream

Evidence Back

Appendix - Selection on Gains: Evidence

	(1) Log final dose	(2) Log grant dollars upstream	(3) Log eligible cost upstream
Panel A. The circle's pre-program abundance			
Level, per standard deviation	0.54***	−0.080	−0.084
	(0.15)	(0.10)	(0.10)
Trend, per standard deviation	−0.30***	−0.066	−0.047
	(0.11)	(0.068)	(0.067)
Joint p-value, level and trend	below 0.001	0.38	0.49
Panel B. The circle's pre-program richness			
Level, per standard deviation	0.61***	0.093	0.12
	(0.18)	(0.11)	(0.11)
Trend, per standard deviation	0.012	0.12	0.12
	(0.11)	(0.077)	(0.078)
Joint p-value, level and trend	0.0024	0.26	0.22
Circles	453	474	474
Pre-program effort, largest river's flow and drainage area	Yes	Yes	Yes

Least squares, one row per treated circle. Robust standard errors in parentheses. *** p-value below 0.01. Circles with more birds before the program ended with larger doses, but not with more grant dollars.

Back

Appendix - Technology: Capacity-Only Commissionings

Question: do commissionings that only added capacity, and so cleaned nothing, move the birds?

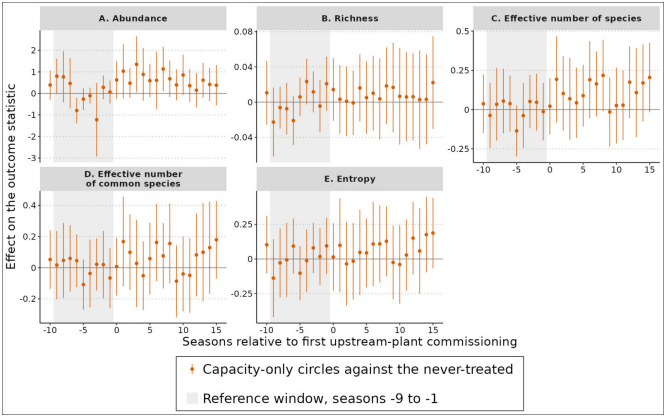
$$\mathbb{E}[\,N_{ct} \mid \cdot\,] \;=\; \phi_c \exp\Bigg(\lambda_t \;+\; \sum_{k\neq -1} \underbrace{\beta_k^{\text{cap}}}_{\substack{\text{effect } k \text{ years after a} \\ \text{capacity-only commissioning}}} \mathbf{1}[\,t - G_c = k\,] \;+\; \underbrace{\mathcal{C}'_{ct}\,\delta + \mathcal{T}_{ct}}_{\substack{\text{controls,} \\ \text{baseline trends}}}\Bigg) \tag{8}$$

where,

- Sample: the circles whose largest commissioned share came from such a plant, and the never-treated
- The test: $\beta_k^{\text{cap}} = 0$ after commissioning

Evidence Back

Appendix - Technology: Evidence



All birds, the main specification. These circles counted 97% more birds per survey after commissioning (0.68 log points, SE 0.36, pre-trend p-value 0.06). Their water moved too. Thus, the comparison is no placebo.

Back

Appendix - The Plant as a Place: Plants Off the Outlet

Question: do plants inside a circle whose effluent never reaches its outlet move its birds?

$$\mathbb{E}[\, N_{ct} \mid \cdot \,] \;=\; \phi_c \exp\Bigg(\lambda_t \;+\; \underbrace{\nu}_{\substack{\text{effect} \\ \text{per plant}}} \underbrace{n^{\mathrm{nd}}_{ct}}_{\substack{\text{plants inside, not} \\ \text{draining to the outlet}}} \;+\; \underbrace{\gamma\, D_{ct}}_{\text{the dose}} \;+\; \underbrace{\mathcal{C}'_{ct}\,\delta + \mathcal{T}_{ct}}_{\substack{\text{controls,} \\ \text{baseline trends}}} \Bigg) \tag{9}$$

where,

- The test: $\nu = 0$, since these plants' effluent never reaches the circle's water
- Beside it: plants with no route to surface water

Evidence Back

Appendix - The Plant as a Place: Evidence

	(1) Not draining to the outlet	(2) No route to surface water
Panel A. All birds		
Abundance, log points per plant	0.31**	0.28
	(0.14)	(0.21)
Effective number of species, SD per plant	0.087***	0.098*
	(0.033)	(0.053)
Surveys	86,860	85,930
Panel B. Waterfowl		
Abundance, log points per plant	−0.18**	−0.44**
	(0.075)	(0.18)
Effective number of species, SD per plant	0.12***	0.057
	(0.031)	(0.059)
Surveys	86,060	85,140
Circles with such a plant inside	126	260
Dose, controls, and trends	Yes	Yes
Circle and year effects	Yes	Yes

Each estimate: the count of such plants inside the circle, beside the dose. Poisson for abundance, least squares for the effective number. Standard errors clustered by circle. ***, **, *: p-values below 0.01, 0.05, 0.10. Waterfowl fall where such plants sit.

Back

Appendix - Number of Plants: The Regression

Specification: equation (3) with the number of plants upstream in place of the dose

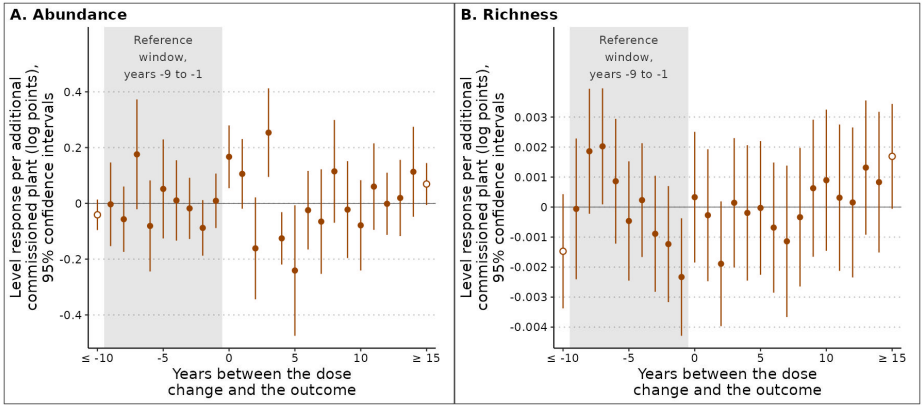
$$\mathbb{E}[\, N_{ct} \mid \mathbf{x}_c, \phi_c \,] \;=\; \phi_c \exp\Bigg(\lambda_t \;+\; \sum_{\tau \neq -1} \underbrace{\varphi_\tau}_{\substack{\text{response } \tau \text{ years} \\ \text{after one more plant}}} \underbrace{\Delta P_{c,t-\tau}}_{\substack{\text{change in the plants} \\ \text{upstream, } \tau \text{ years earlier}}} \;+\; \mathcal{C}'_{ct}\,\delta + \mathcal{T}_{ct} \Bigg) \tag{10}$$

where,

- P_{ct}: the grant plants commissioned within 25 river-miles upstream of circle c by year t, the dose with every share set to one
- The pre-trend test does not reject in 68 of 85 regressions

The Result Back

Appendix - Number of Plants: The Result



All birds, the main specification. Fourteen years on, each added plant raises richness by 0.0032 log points (SE 0.0014). For the water-linked species it is 0.0096 (SE 0.0029), about 1% more species per survey, and for landbirds 0.0002 (SE 0.0014).

Back

Appendix - Water Quality After Commissioning: Evidence

Panel A. All treated circles

	(1) Effect of commissioning, standard deviations	(2) Pre-trend test p-value	(3) Circle-years
Regulated			
Biochemical oxygen demand	0.047	0.62	27,449
	(0.050)		
Total suspended solids	−0.034	0.31	66,153
	(0.043)		
Fecal coliform, log	0.00073	0.74	45,643
	(0.034)		
Dissolved oxygen saturation	0.15**	0.06	36,401
	(0.066)		
Partly regulated			
Total phosphorus	−0.15**	0.20	57,804
	(0.066)		
Nitrogen			
Ammonia	0.082	0.04	53,583
	(0.062)		
Total nitrogen	0.053	0.86	47,188
	(0.045)		
Nitrate plus nitrite	0.038	0.13	43,889
	(0.069)		
Falsification			
Lead (Pb)	0.041		43,859

Appendix - Predicted Cleaning: Evidence

	(1) From the plants' cut in suspended solids	(2) From the halves split on the plants' suspended solids before the grants	(3) From the halves split on the plants' treatment level, on fecal coliform
Panel A. Abundance, Poisson quasi-likelihood, in log points			
Dose alone	0.0067*	0.0068*	0.0082
	(0.0039)	(0.0038)	(0.0050)
Dose, with predicted cleaning	0.0061	0.0073	0.020*
	(0.0039)	(0.0067)	(0.012)
Predicted cleaning	−0.012*	0.017	−1.5
	(0.0064)	(0.13)	(1.9)
Surveys	79,120	82,810	81,950
Panel B. Richness, Poisson quasi-likelihood, in log points			
Dose alone	−0.00025***	−0.00021**	−0.00017
	(0.000089)	(0.00010)	(0.00014)
Dose, with predicted cleaning	−0.00025***	−0.00036	−0.0011**
	(0.000088)	(0.00023)	(0.00042)
Predicted cleaning	−0.00013	−0.0051	0.12**
	(0.00036)	(0.0061)	(0.057)
Surveys	79,120	82,810	81,950
Panel C. Effective number of species, least squares, in standard deviations			
Dose alone	0.0056***	0.0055***	0.0044**
	(0.0015)	(0.0015)	(0.0020)
Dose, with predicted cleaning	0.0063***	0.0045**	−0.00028
	(0.0012)	(0.0021)	(0.0028)
Predicted cleaning	0.012	−0.039	0.61***
	(0.011)	(0.065)	(0.24)
Surveys	79,120	82,810	81,950
Circle and year effects	Yes	Yes	Yes
Effort controls	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes
Attribute-by-year trends	Yes	Yes	Yes

Each column is one static regression of the statistic on the dose and predicted cleaning, with circle and year effects and the main controls. Standard errors clustered by circle. The event-study versions fail their pre-trend tests.

Back

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